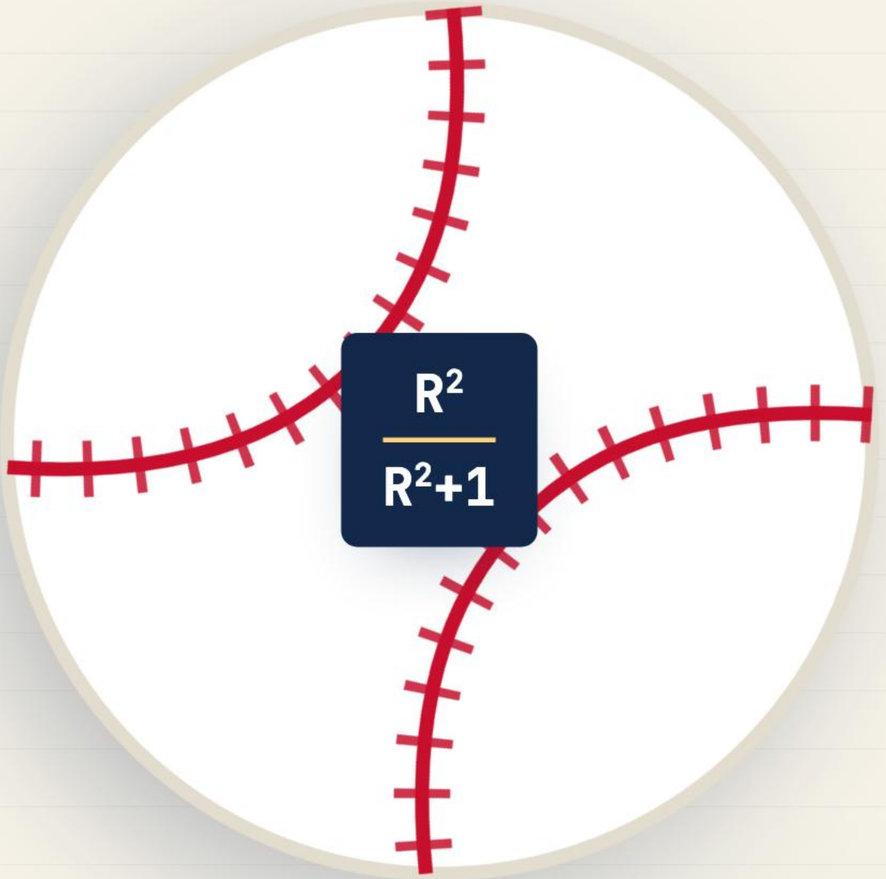
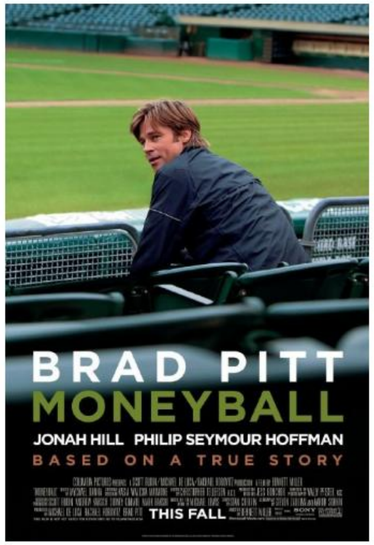
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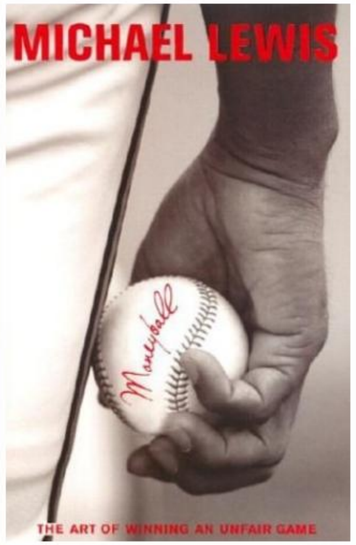
BASEBALL ANALYTICS:

MATHLETICS CH.1

SPM261 Intro to Sport Management









SABERMETRICS

http://m.mlb.com/glossary/advanced-stats









https://www.youtube.com/watch?v=JXyisZLObHc&list=PLZbXA4lyCtqosSLgdOSwUWDK_KH9c2rIq



https://www.youtube.com/watch?v=aNDj-H1jxV0&list=PLZbXA4lyCtqosSLgdOSwUWDK_KH9c2rIq&index=2



WHAT IS SPORT ANALYTICS?

01 Data

02 Analyzing skill

03 Data interpretation

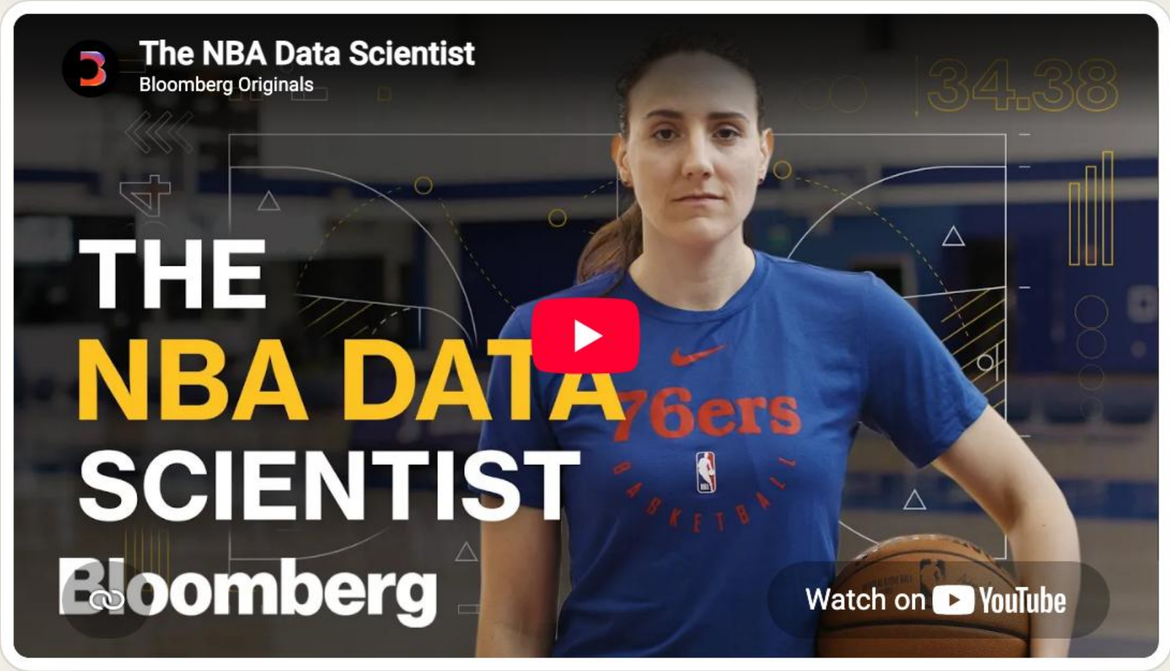
04 Basic features of the data

05 Data visualization

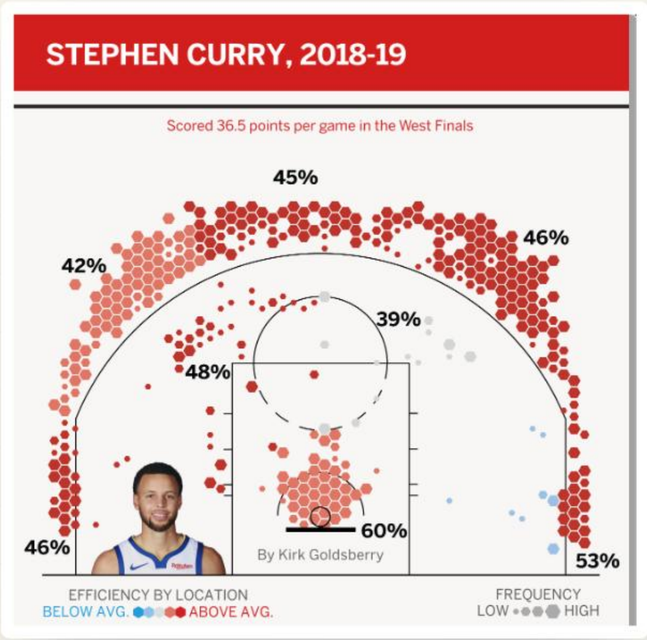
06 Prediction-significance test

WHAT IS SPORT ANALYTICS? CONT.

Player/team performance, winning/ performance prediction, fan emotions/reactions (i.e., social media), economic impact, betting market, rule/ policy test, playoff team selection, etc.



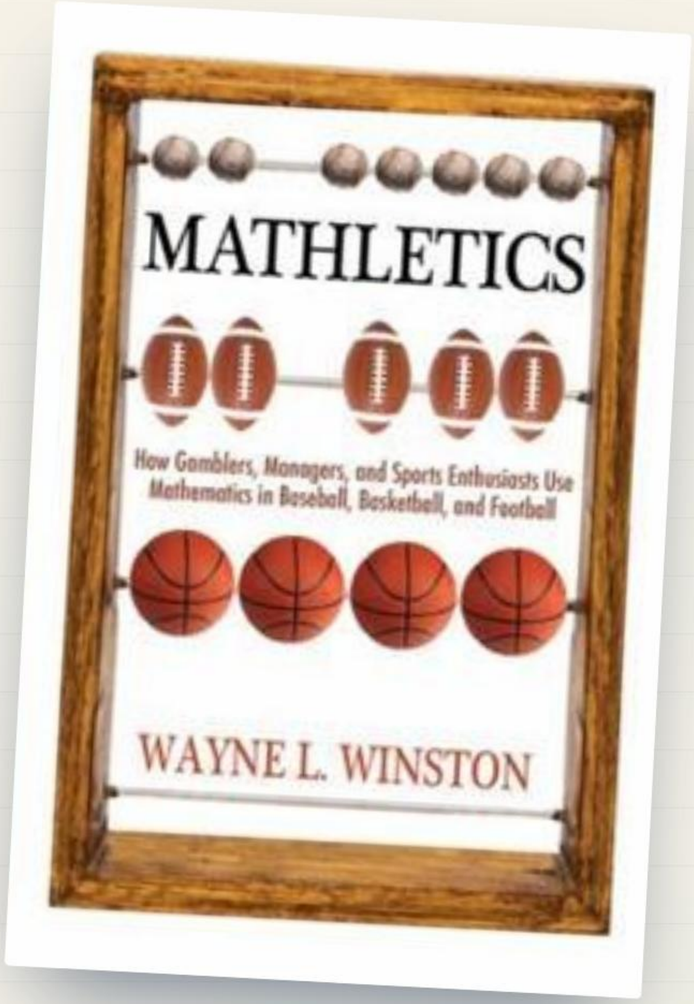
https://www.youtube.com/watch?v=MpLHMKTolVw



PLEASE READ THE BOOK MATHLETICS:

(a) Ch.1 Baseball's Pythagorean Theorem

You can download the book Mathletics on course website. See Module 10





MATHLETICS · CHAPTER 1

BASEBALL'S PYTHAGOREAN THEOREM (CH.1)



PREDICTING A TEAM'S WINNING PERCENTAGE & # OF GAMES

BASEBALL'S PYTHAGOREAN THEOREM (CH.1) CONT.

- The more runs a baseball team scores, the more games the team should win.
- Bill James introduces a formula that measures percentage of games won by baseball team (p.3).
- Pythagorean Theorem states that a triangle is a right triangle if and only if $a^2+b^2=c^2$.
- For example, a triangle with sides of lengths 3, 4, and 5 is right triangle because $3^2+4^2=5^2$.

$$\frac{\text{runs scored}^2}{\text{runs scored}^2 + \text{runs allowed}^2} = \text{estimate of percentage of games won.} \quad (1)$$

This formula has several desirable properties.

- The predicted win percentage is always between 0 and 1.
- An increase in runs scored increases predicted win percentage.
- A decrease in runs allowed increases predicted win percentage.

If R (scoring ratio) = $\frac{\text{runs scored}}{\text{runs allowed}}$

The equation (1) can be interpreted as

$$\frac{R^2}{R^2+1} = \text{estimate of percentage of games won}$$

BASEBALL'S PYTHAGOREAN THEOREM (CH.1) CONT.

WHY?

The equation (1) is same as the following fraction (if you divide the numerator and denominator of (runs allowed)^2:

$$\frac{\frac{\text{runs scored}^2}{\text{runs allowed}^2}}{\frac{\text{runs score}^2}{\text{runs allowed}^2} + \frac{\text{runs allowed}^2}{\text{runs allowed}^2}}$$

If R (scoring ratio) = $\frac{\text{runs scored}}{\text{runs allowed}}$

$$\frac{R^2}{R^2 + 1}$$ **= estimate of percentage of games won**

BASEBALL'S PYTHAGOREAN THEOREM (CH.1) CONT.

Figure 1.1 shows how well $\frac{R^2}{R^2+1}$ predicts MLB team's winning percentage.

1. For example, the 2006 Detroit Tigers scored 822 runs and gave up 675 runs.
2. Their scoring ratio was R (scoring ratio) = $\frac{822}{675}$ = **1.218**
3. Their predicted win percentage was $\frac{1.218^2}{1.218^2+1}$ = **.597**
4. Actual win percentage (win/total games played) was **.586**
5. Absolute error (actual win % - predicted win %)
   So Tigers absolute error was |.586-.597| = **0.011**

MAD = 0.020

Year	Team	Wins	Losses	Runs scored	Runs allowed	Scoring ratio	Predicted winning %	Actual Winning %	Absolute Error
2006	Diamondbacks	76	86	773	788	0.981	0.490	0.469	0.021
2006	Braves	79	83	849	805	1.055	0.527	0.488	0.039
2006	Orioles	70	92	768	899	0.854	0.422	0.432	0.010
2006	Red Sox	86	76	820	825	0.994	0.497	0.531	0.034
2006	White Sox	90	72	868	794	1.093	0.544	0.556	0.011
2006	Cubs	66	96	716	834	0.859	0.424	0.407	0.017
2006	Reds	80	82	749	801	0.935	0.466	0.494	0.027
2006	Indians	78	84	870	782	1.113	0.553	0.481	0.072
2006	Rockies	76	86	813	812	1.001	0.501	0.469	0.031
2006	**Tigers**	**95**	**67**	**822**	**675**	**1.218**	**0.597**	**0.586**	**0.011**
2006	Marlins	78	84	758	772	0.982	0.491	0.481	0.009
2006	Astros	82	80	735	719	1.022	0.511	0.506	0.005
2006	Royals	62	100	757	971	0.780	0.378	0.383	0.005
2006	Angels	89	73	766	732	1.046	0.523	0.549	0.027
2006	Dodgers	88	74	820	751	1.092	0.544	0.543	0.001
2006	Brewers	75	87	730	833	0.876	0.434	0.463	0.029
2006	Twins	96	66	801	683	1.173	0.579	0.593	0.014
2006	Yankees	97	65	930	767	1.213	0.595	0.599	0.004

Figure 1.1. Baseball's Pythagorean Theorem, 1980–2006. See file Standings.xls.

EXERCISE

EXERCISE

2026 MLB_Student Excel file on course website (see Module 10) or link below

https://docs.google.com/spreadsheets/d/1cKcHZ_6WceI8Xxjtd9ODp5I-pTDoEIzeLjIoWplFZUI/edit?usp=sharing

Complete the following columns: Scoring Ratio, Predicted Winning %, Actual Winning %, Absolute Error

SCORING RATIO	PREDICTED WINNING %	ACTUAL WINNING %	ABSOLUTE ERROR

- Upload your completed answer to Module 10 (see In-Class Assignments- Baseball Analytics)
- This will replace your handwritten note.

IN-CLASS ASSIGNMENT

IN-CLASS ASSIGNMENTS

HOW DO WE USE THIS INFORMATION?

On July 4, 2005, the Nationals were in first place with a record of **50-32**

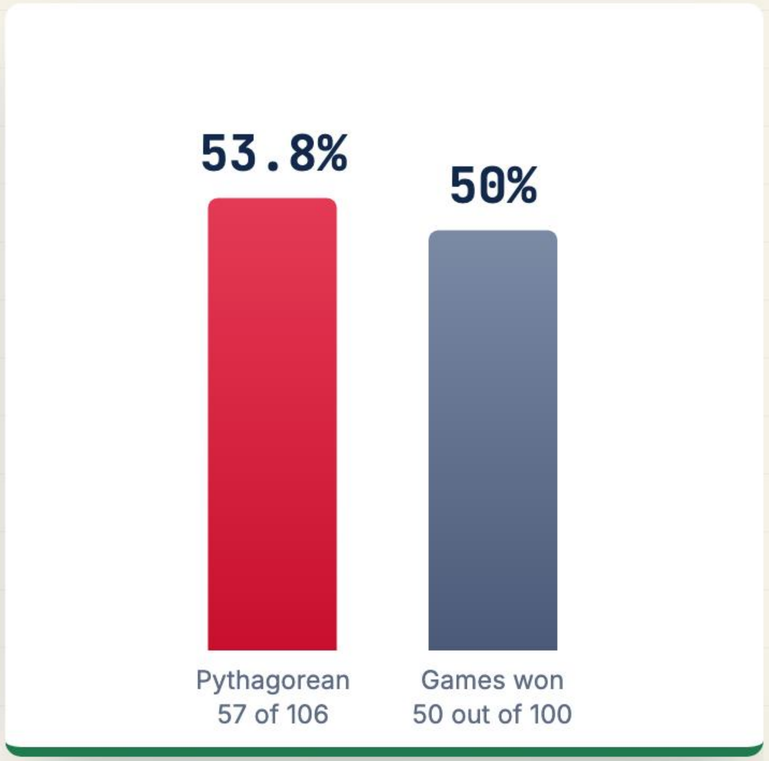
On July 4, 2005, the Nationals scoring ratio was **.991**

- Calculate the 2005 Nationals' predicted season records (# of wins and losses) using the scoring ratio. MLB has a total of 162 games per season.
- In 2005, National finished XX (# of win) – XX (# of loss)?
- This will replace your handwritten note.
- Please download the form (from Module 10) and return your answer with the formula.

BASEBALL'S PYTHAGOREAN THEOREM (CH.1) CONT.

HOW WELL DOES THE PYTHAGOREAN THEOREM FORECAST?

- Compared the Pythagorean Theorem's forecast for each MLB playoff series (1980-2007) against a prediction based just on games won.
- For each playoff series the Pythagorean method would predict the winner to be the team with the higher scoring ratio, while the games won approach simply predicts the winner of a playoff series to be the team that won more games.
- Pythagorean approach correctly predicted 57 of 106 playoff series (53.8%), while the games won approach correctly predicted the winner of only 50% (50 out of 100) playoff series.
- Regular season is a relatively poor predictor of the playoffs in baseball because a team's regular season record depends on the performance of five starting pitchers.
- During the playoffs teams only use three or four starting pitchers, so much of the regular season data are not relevant for predicting the outcome of the playoffs.





QUESTIONS?

SPM 261 · BASEBALL ANALYTICS